The Self as a Dynamical Attractor: Emergence of Self-Like Dynamics in Spiking Neural Networks under Continuous Embodied Input

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Abstract

The nature of the self is traditionally modeled as a persistent internal state, memory structure, or representational object. However, empirical phenomena such as the temporary disappearance of subjectivity during deep sleep or general anesthesia challenge models that require continuous preservation of specific neural states.

In this work we propose an alternative hypothesis: the self emerges as a dynamical attractor of neural activity rather than as a stored internal representation.

Using spiking neural network simulations implemented in the Brian2 framework, we show that self-like functional properties can arise from slow adaptive dynamics driven by continuous endogenous input. A recursive observer layer introduces predictive modulation that alters but does not generate the underlying dynamical regime.

We further introduce a recursive architecture in which slow embodied dynamics interact with a faster observer layer that generates predictive modulation. A parameter sweep of prediction gain reveals that recursive prediction modulates but does not generate the underlying attractor dynamics.

These results support a view of the self as a transient dynamical structure emerging from the interaction between embodied input, neural adaptation, and recursive prediction.

1 Introduction

The nature of the self remains one of the central questions in cognitive science and philosophy of mind. Many classical models implicitly treat the self as a persistent internal entity, often associated with stable neural representations, memory structures, or centralized control mechanisms.

However, empirical observations challenge the necessity of persistent internal representations. Human subjectivity can temporarily disappear during deep sleep, general anesthesia, or certain neurological states, yet subjective experience can later re-emerge when neural activity resumes. These phenomena suggest that the functional properties associated with the self may depend primarily on the dynamical regime of neural activity rather than on the continuous preservation of a specific internal state.

Recent developments in theoretical neuroscience increasingly emphasize dynamical systems approaches to cognition. In such frameworks, cognitive phenomena emerge from the collective dynamics of neural networks interacting with external inputs and internal constraints. Within this perspective, stable functional structures can arise as attractors of neural dynamics rather than as stored objects.

In the present work we explore the hypothesis that self-like functional properties emerge as dynamical attractors in adaptive neural systems. Specifically, we investigate whether a neural network receiving continuous endogenous input can develop stable dynamical regimes that persist despite stochastic perturbations and changes in external inputs.

To examine this hypothesis we implement a spiking neural network model with two interacting dynamical layers. A slow layer integrates endogenous signals representing continuous embodied input, while a faster observer layer introduces predictive modulation. We demonstrate through computational experiments that organized network dynamics can arise from the interaction between these layers, and that recursive prediction modifies but does not create the underlying attractor structure.

We evaluate this framework through a series of computational experiments examining three aspects of the model dynamics: (i) the role of continuous embodied input in stabilizing network activity, (ii) the influence of the observer layer on collective neural dynamics, and (iii) the effect of recursive prediction on the resulting dynamical regime. These experiments test whether self-like functional properties can arise as stable attractors of neural dynamics under continuous endogenous input.

2 Theoretical Framework

Biological organisms continuously receive endogenous signals originating from bodily processes. These include proprioceptive, visceral, and autonomic signals that are present even in the absence of external sensory stimuli. A substantial contribution to this continuous background activity arises from the enteric nervous system, which operates with a high degree of autonomy and generates ongoing rhythmic and stochastic neural activity within the gastrointestinal system. Together, these processes provide a persistent stream of endogenous signals with relatively stable statistical properties.

From the perspective of dynamical systems, such signals may provide a stable background input that allows neural networks to develop persistent dynamical regimes. Rather than constructing the self from discrete sensory events, the system may instead stabilize a dynamical structure in response to the continuous statistical structure of embodied input.

We therefore distinguish two interacting dynamical components:

$$\text{Slow embodied dynamics}(\Sigma_{slow})$$

A neural population integrating endogenous signals with slow adaptive dynamics.

$$\text{Fast observer dynamics}(\Sigma_{fast})$$

A faster layer that responds to sensory patterns and generates modulatory signals.

These components interact through a recursive loop in which slow network activity influences fast observer dynamics, which in turn modulates the slow layer.

Conceptually the architecture can be represented as

$$\text{body noise} \rightarrow \Sigma_{slow} \rightarrow \text{self dynamics}$$

$$\Sigma_{slow} \rightarrow \Sigma_{fast} \rightarrow \text{prediction}$$

$$\text{prediction} \rightarrow \text{correction} \rightarrow \Sigma_{slow}$$

Within this framework the self is not treated as a stored representation but as a stable dynamical regime arising from slow adaptive neural activity.

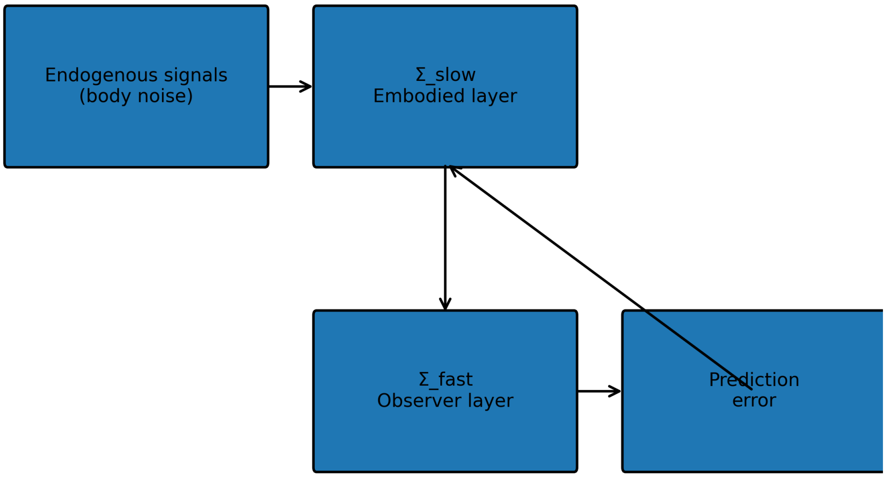

Figure 1: The system consists of a slow embodied neural layer (Σ_{slow}) receiving continuous endogenous input representing bodily signals. A faster observer layer (Σ_{fast}) responds to temporal patterns and generates predictive signals. Prediction error modulates the slow layer through a recursive feedback loop.

3 Model Architecture

The model consists of two interacting populations of adaptive spiking neurons implemented using the Brian2 simulation framework.

3.1 Slow embodied layer (Σ_{slow})

The slow layer contains adaptive spiking neurons with membrane dynamics governed by

$$\frac{dv}{dt} = \frac{E_L - v + R(I_{endo} + I_{mod} - w)}{\tau_m} \tag{1}$$

where v is the membrane potential and I_{endo} represents endogenous input modeled as an Ornstein–Uhlenbeck process.

The adaptation variable evolves according to

$$\frac{dw}{dt} = \frac{a(v - E_L) - w}{\tau_w} \tag{2}$$

Endogenous input is modeled as

$$\frac{dI_{endo}}{dt} = -\frac{(I_{endo} - \mu)}{\tau} + \sigma\sqrt{\frac{2}{\tau}}\xi(t) \tag{3}$$

where $\xi(t)$ is Gaussian white noise.

3.2 Fast observer layer (Σ_{fast})

The fast layer consists of neurons with shorter membrane time constants that respond to temporal input patterns and generate modulatory signals affecting the slow layer. The activity of the observer layer is summarized by a slowly decaying population variable representing recent spiking activity.

3.3 Predictive modulation

Prediction is implemented by comparing the current network state with a running estimate:

$$I_{pred} = -g_{pred}(x - \hat{x}) \tag{4}$$

where g_{pred} is the prediction gain.

4 Computational Experiments

Several computational experiments were performed to investigate the dynamical properties of the system.

4.1 Embodied input

Three conditions were tested:

- endogenous input only
- no endogenous input
- endogenous input with external modulation

This experiment tests whether continuous embodied input is necessary for the emergence of organized network dynamics.

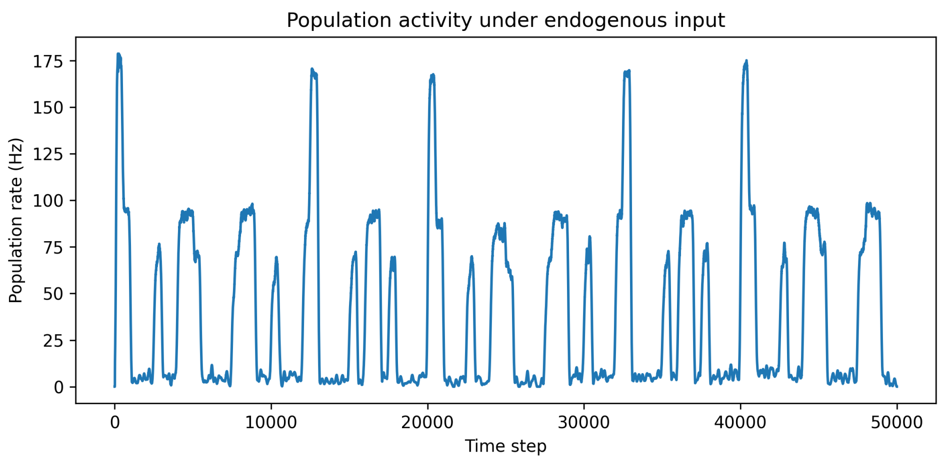

Figure 2: Population activity under continuous endogenous input.

4.2 Observer modulation

The observer layer was introduced to examine how fast neural dynamics influence the stability of the slow dynamical regime. The goal was to examine whether the observer layer stabilizes or destabilizes the dynamical regime of the slow layer.

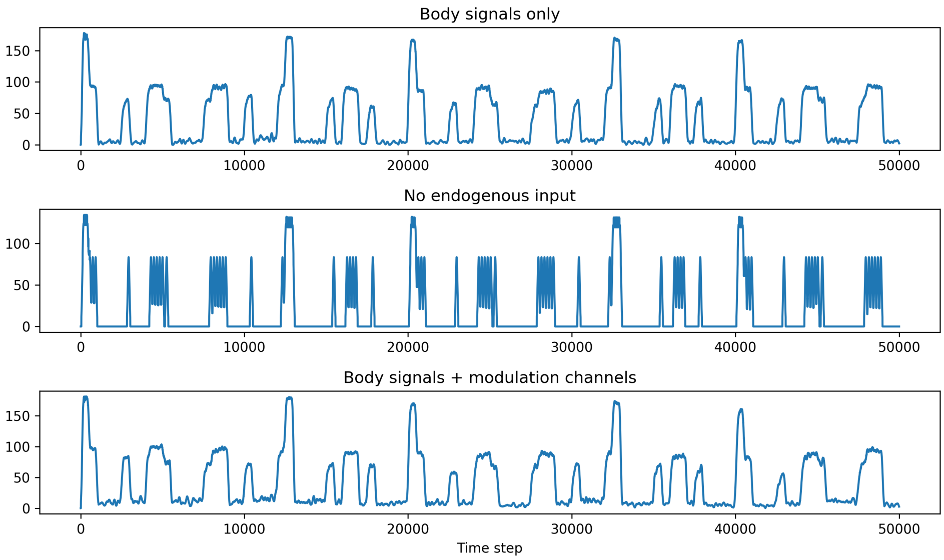

Figure 3: Body signals + modulation channel.

4.3 Recursive prediction

A recursive feedback loop was implemented in which prediction error modulates the slow neural population. This configuration implements a minimal form of self-modeling dynamics.

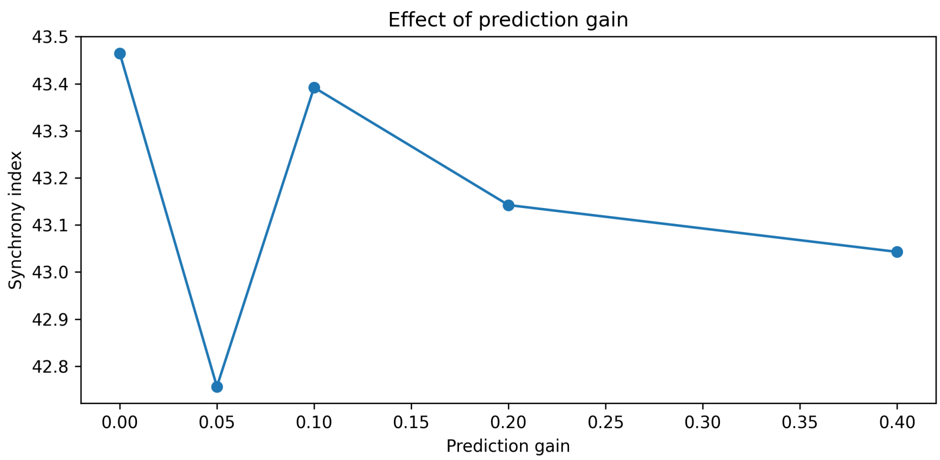

Figure 4: Effect of prediction gain.

4.4 Prediction gain sweep

Finally, we varied the prediction gain parameter across a range of values to evaluate its effect on network dynamics. This experiment tests whether recursive prediction generates or merely modulates the underlying attractor structure.

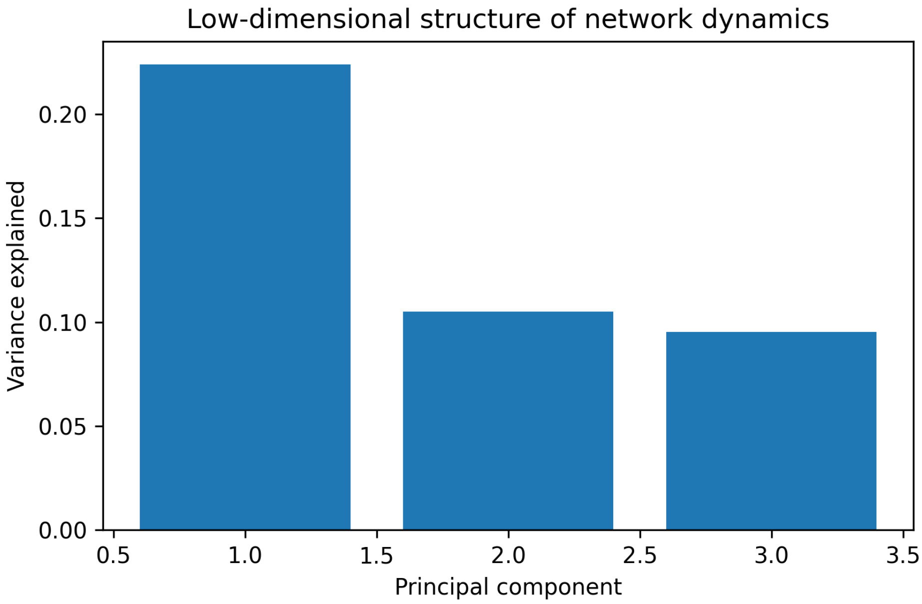

Figure 5: Low-dimensional structure of network dynamics.

5 Results

Across simulations the slow neural population exhibited organized collective dynamics when endogenous input was present. Removing the endogenous drive significantly reduced structured network activity, indicating that continuous embodied signals play a critical role in stabilizing the dynamical regime.

Introducing the observer layer altered the statistical properties of network activity but did not fundamentally change the presence of the attractor-like dynamics.

The recursive prediction loop modulated the stability of the network state without generating the dynamical regime itself. Parameter sweeps of prediction gain showed gradual changes in spike rate and network variance but did not produce qualitative phase transitions.

Principal component analysis of membrane potentials revealed low-dimensional structure in the network dynamics, suggesting that the collective behavior of the network is governed by a small number of dominant dynamical modes.

6 Discussion

The results support the hypothesis that self-like functional properties can emerge as dynamical attractors in adaptive neural systems receiving continuous embodied input.

Rather than being stored as persistent internal representations, such structures arise from the interaction between slow adaptive dynamics, recurrent connectivity, and statistically stable background input.

The recursive predictive loop modulates and stabilizes but does not create these dynamics, suggesting that predictive processing may operate on top of a deeper dynamical structure generated by embodied neural activity.

This perspective is consistent with dynamical systems approaches to cognition in which stable behavioral and cognitive patterns arise as attractors of neural dynamics.

7 Methods

Simulations were implemented using the Brian2 spiking neural network simulator. Neural populations consisted of adaptive integrate-and-fire neurons with stochastic Ornstein–Uhlenbeck input processes.

Network activity was analyzed using spike statistics, population synchrony measures, autocorrelation analysis, and principal component analysis of membrane potentials.

Code Availability

All simulation code used in this study is available at

`https://github.com/dlevin-coder/protoself`

A versioned archive of the code and experimental data is available through Zenodo.

References

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